

Task - 14: Linux Server Hardening & Secure Configuration

Internship Task – Elevate Labs

This task focused on securing a Linux server by applying standard hardening techniques. The objective was to reduce the attack surface, protect system resources, and ensure the server is resistant to common cyber attacks.

1. Reviewing Default Linux System Settings

The first step was to review default system configurations. This included checking existing users, running services, and open network ports. Understanding the default state helps identify unnecessary exposure.

2. User Account Management

Unused user accounts were identified and removed. Sudo access was restricted by following the principle of least privilege, ensuring only required users had administrative permissions.

3. Disabling Root Login & Securing SSH

Direct root login was disabled to prevent brute-force attacks. SSH was configured to use key-based authentication instead of passwords, improving remote access security.

4. System Updates & Patch Management

System packages were updated regularly to fix known vulnerabilities. Automatic security updates were enabled to ensure timely protection against newly discovered threats.

5. Firewall Configuration

A firewall was configured to allow only required network traffic. Unnecessary ports were blocked, reducing the risk of unauthorized access.

6. Disabling Unnecessary Services

Running services were reviewed and unused services were stopped and disabled. Reducing active services minimizes potential entry points for attackers.

7. Securing File Permissions

Sensitive system and configuration files were secured using proper file permissions. This prevented unauthorized users from modifying critical system files.

8. Log Monitoring

System and authentication logs were reviewed to monitor user activity. Log analysis helps in detecting suspicious behavior and potential security incidents.

Security Configuration Summary

- Removed unused user accounts - Restricted sudo access - Disabled root login - Enabled SSH key-based authentication - Updated system packages - Configured firewall rules - Disabled unnecessary services - Secured file permissions - Reviewed system logs

Learning Outcome

Through this task, I gained practical experience in Linux server hardening. I learned how to apply secure configurations, reduce attack surfaces, and protect Linux systems against common security threats.